Definition : Thresholding is an operation which is used to replace costly (resource intensive) floating point operations by a cheaper comparison operation for binary neural networks. In the context of multi-bit quantization (INT2,INT3….INT8), this operation is referred to as multi-thresholding. (as multiple thresholds are required to represent those operations.) Here, the thresholds themselves can be floating point numbers. There is no constraint n them being INT datatype.

This operation enables the synthesis of very fast implementations of neural networks on FPGAs.

Condition : This operation can be used to replace only monotonically increasing functions. If this property is not satisfied by the function then we cannot convert that operation into Mutli-thresholding operation.

Working example : We see an example of (Sigmoid + QuantLinear) op conversion to Multithresholding operation. We consider the case of INT4 quantization.

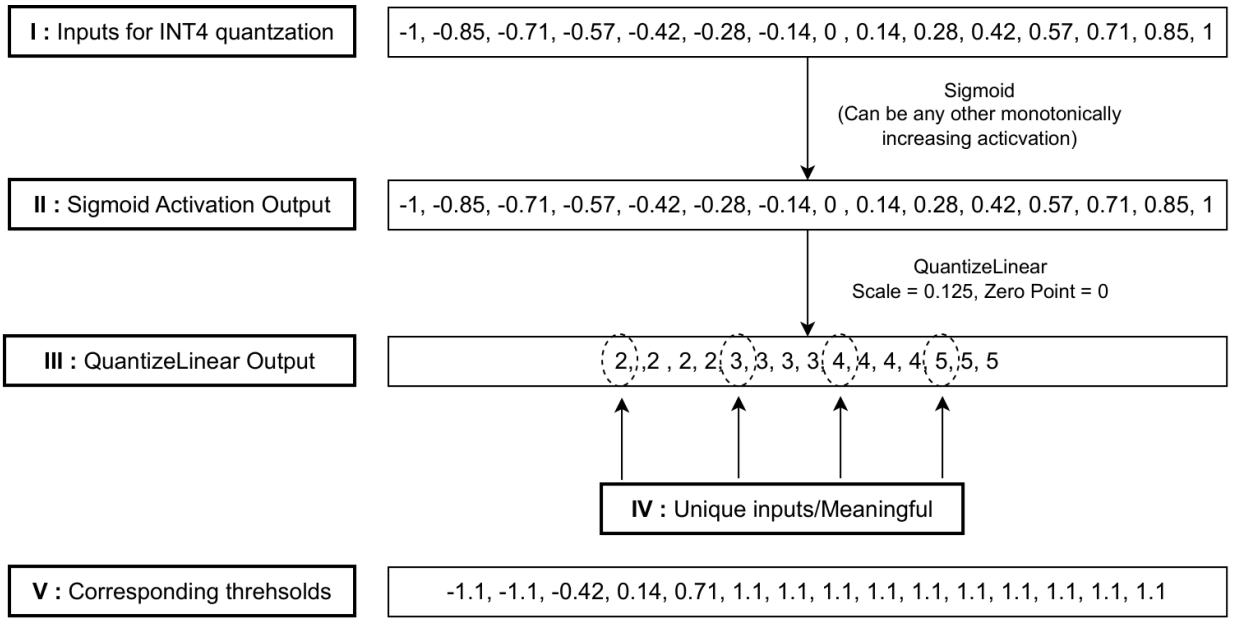
Assumptions :

1. The input the sigmoid function gets is in the range [-1,1]
2. Quantization scale is 0.125 ; Zero-point is 0.

Steps :

1. Inputs for which the thresholds needs to be generated.
2. Apply Sigmoid function on that.
3. Apply the QuantizeLinear operation on that output.
4. Identify unique inputs for which the outputs change. These unique inputs will be our thresholds.
5. For loop which identifies what the thresholds should be. The working procedure in detail is demonstrated in the bottom figure.

Example Steps :



NOTE : In the above process, we convert the *Sigmoid* + *QuantizeLinear* operation with *Multithrehsolding* operation with the above calculated thersholds.